

Cross-Disciplinary Teaching in Vocational Colleges: A Self-Directed Model for Teacher Professional Growth

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Abstract

Cross-disciplinary teaching assignments represent a systemic challenge in vocational education worldwide, particularly in developing economies where specialist teacher shortages intersect with rapid curricular expansion. This paper reports a longitudinal case study of a vocational teacher whose multi-convergent UAV practice trajectory is traced from undergraduate study to the present, spanning over a decade. Analytical focus falls on the most recent four-year period (2021–2025), during which a self-directed, technology-assisted model enabled the development of cross-disciplinary teaching competence. Teacher Z (a pseudonym) held no formal qualification in UAV technology, yet through the Narrative-Framework-Scene-Creation (NFSC) methodology, supported by a multi-agent software system, developed competence across 10 UAV courses and secured three funded research projects together with a copyright registration.

Drawing on teaching journals (2021–2025), system usage records, and these institutional outputs, the study traces the teacher's trajectory from disciplinary novice to research-active academic. Findings indicate that the self-directed model supported three developmental pathways: knowledge domain mapping across disciplines, progressive curriculum design, and the integration of teaching practice with scholarly output. Three auxiliary cases—a solid-waste-to-smart-engineering transition and a 60-year-old engineer's self-directed study of traditional music—offer preliminary evidence of transferability across disciplines and life stages. The paper argues that this zero-cost, self-directable model holds significant policy implications for teacher training in emerging economies, particularly within South-South cooperation frameworks such as BRICS. The model is positioned not as a replacement for formal teacher education but as a pragmatic bridge in settings where formal training remains inaccessible.

****Keywords:**** cross-disciplinary teaching; self-directed professional development; vocational education; technology-assisted learning; teacher professional growth; scarce-resource contexts; BRICS education cooperation

§1 Introduction

The global expansion of vocational education has created a persistent structural problem: specialist teacher shortages force institutions to assign instructors to teach outside their disciplinary training. This is not a matter of isolated cases or individual career choices but a systemic consequence of policy-driven enrolment growth outpacing the supply of trained teachers (Organisation for Economic Co-operation and Development [OECD], 2022). The World Bank (2023) identifies skill mismatches in vocational teaching as a binding constraint on education quality in middle-income countries, where rapid industrial upgrading generates demand for new curricula faster than teacher education systems can respond.

China illustrates this tension with particular clarity. Following the State Council's *National Vocational Education Reform Implementation Plan* (2019), vocational college enrolments expanded by over 30 per cent within five years, creating demand for instructors in emerging fields—artificial intelligence applications, renewable energy systems, drone technology—for which no established teacher training pathway existed (Koo, 2016). Colleges responded by reassigning existing faculty to these new programmes, producing a cohort of cross-disciplinary

teachers whose formal qualifications were in other fields. These teachers face a dual burden: acquiring new disciplinary knowledge while simultaneously maintaining teaching quality in programmes that lead to professional certification for their students.

Traditional approaches to teacher professional development are poorly suited to this challenge. Off-site training programmes are costly and logistically unsustainable at scale (Desimone, 2009). Degree study—returning to university for a second bachelor's or master's degree—requires time and financial resources unavailable to working teachers in under-resourced institutions (Darling-Hammond et al., 2017). School-based research groups, a well-established feature of Chinese teacher culture, typically assume disciplinary homogeneity and offer limited support to the lone cross-disciplinary teacher in a department (Cochran-Smith & Lytle, 2001). Consequently, thousands of vocational teachers are left to bridge cross-disciplinary gaps on their own, with little institutional or scholarly guidance.

Self-directed learning, as theorised by Knowles (1975) and developed by Brookfield (1986), offers a conceptual foundation for addressing this gap. When combined with contemporary information technologies functioning as always-available knowledge scaffolds, such a model may enable teachers to construct disciplinary competence without formal training. Whether this potential translates into actual professional growth in cross-disciplinary vocational settings remains largely undocumented.

This paper reports a longitudinal case study of Teacher Z, a vocational college instructor at [University A] in China. The study traces a multi-convergent UAV practice trajectory spanning over a decade from undergraduate studies, with analytical focus on the most recent four years (2021–2025), during which a self-directed, technology-assisted model enabled the transition to teaching competence. Teacher Z holds a bachelor's degree in land resource management and a master's degree in environmental engineering—neither in UAV technology—yet has, since 2021, built a UAV programme, secured three funded research projects in 2025, and registered intellectual property for his teaching system. The teacher achieved this through a self-directed learning model: the Narrative-Framework-Scene-Creation (NFSC) approach, supported by a multi-agent software system deployed at zero additional cost to the institution.

The study addresses two research questions:

- **RQ1:** How does a self-directed, technology-assisted learning model support cross-disciplinary teaching competence development in a vocational college context?
- **RQ2:** What conditions enable the transfer of this model across disciplines and institutional contexts?

The paper is structured as follows. Section 2 reviews relevant literature on cross-disciplinary teaching, self-directed teacher learning, and technology-supported professional development. Section 3 describes the longitudinal case study methodology. Section 4 presents findings across three developmental phases. Section 5 discusses the findings with reference to the research questions and draws out policy implications for developing-country contexts, with particular attention to the South African development state and BRICS cooperation frameworks. Section 6 concludes.

§2 Literature Review

2.1 Cross-Disciplinary Teaching in Vocational Education

Cross-disciplinary teaching—the assignment of teachers to subjects outside their formal qualification area—has been documented as a structural feature of education systems under resource constraints (Ingersoll, 2003). In the vocational education sector, the phenomenon is amplified by the sector's obligation to align curricula with rapidly changing labour market demands. When new technologies enter industrial practice, vocational colleges must introduce corresponding programmes, often before a qualified teacher supply has materialised (Billett, 2002).

The theoretical framework of boundary crossing, developed by Akkerman and Bakker (2011), provides a productive lens for examining cross-disciplinary teacher transitions. Boundary crossing refers to the process by which individuals navigate discontinuities between different social or cultural practices—in this case, between the knowledge domains of a teacher's original discipline and their new teaching field. Akkerman and Bakker identify four learning mechanisms activated at boundaries: identification (recognising the boundary), coordination (establishing procedures to bridge it), reflection (developing new perspectives), and transformation (producing new practices that integrate elements from both domains). For cross-disciplinary vocational teachers, successful professional growth requires progress through all four mechanisms.

The vocational education context adds specific complexity. Unlike school-level subject reassignment, vocational cross-disciplinary transitions involve not only content knowledge but also the procedural, equipment-based, and industry-standard practices embedded in technical disciplines (Eraut, 2004). A teacher reassigned from environmental engineering to drone technology, for example, must acquire knowledge spanning aerodynamics, radio communication, remote sensing, image processing, and civil aviation regulations—each a distinct knowledge domain with its own conceptual structure.

2.2 Self-Directed Teacher Learning

Self-directed learning (SDL) has a substantial theoretical lineage. Knowles (1975) defined it as a process in which individuals take the initiative, with or without the help of others, in diagnosing their learning needs, formulating learning goals, identifying resources, choosing and implementing learning strategies, and evaluating learning outcomes. Brookfield (1986) extended this framework by emphasising the social context of self-directed learning and the role of critical reflection in adult learning processes.

In teacher education, SDL has been examined primarily as a complement to formal professional development rather than as a primary mechanism for qualification-level knowledge acquisition (Avalos, 2011). Verster et al. (2024), writing in this journal, demonstrated that self-directed professional development could enhance teachers' curriculum practice when structured as an intentional intervention. Their study of South African teachers showed that those who engaged in self-directed learning activities with structured reflection improved their ability to translate curriculum policy into classroom practice. This finding underscores the potential of SDL approaches but also highlights the importance of structure—self-direction does not mean unguided.

Workplace learning theories provide complementary insights. Eraut (2004) distinguishes between formal learning (planned, structured, credentialised) and informal learning (embedded in work activities, often tacit and unrecognised). For cross-disciplinary teachers, most learning occurs in the informal mode as they prepare lessons, troubleshoot equipment, and respond to student questions. Billett (2002) argues that workplace learning can be effective when the work environment provides appropriate guidance—through access to more knowledgeable colleagues, authentic tasks of graduated difficulty, and direct feedback. The difficulty for the cross-

disciplinary vocational teacher is that the workplace environment lacks precisely these affordances: knowledgeable colleagues in the new discipline are absent, and the task of teaching full courses is inappropriately demanding for a novice.

2.3 Technology-Supported Professional Development

The intersection of technology and teacher professional development has generated a substantial but uneven literature. Clarke and Hollingsworth (2002) established that effective professional development requires sustained engagement, connection to practice, and a reflective component—characteristics that well-designed technology platforms can support. More recently, Naidoo (2026), writing in this journal, assessed ICT integration in South African private secondary schools and found that technology infrastructure alone does not improve teaching outcomes; the critical factor is the alignment of technology use with teachers' identified professional needs.

Rasool and Naidoo (2024) examined ICT sustainability in previously disadvantaged South African public schools and identified cost, maintenance, and ongoing technical support as persistent barriers. Their findings underscore a critical design principle for technology-assisted teacher development in resource-constrained settings: the solution must be zero-cost at the point of use and must require no specialised technical support for ongoing operation. This principle shaped the architecture of the support system used in the present study.

2.4 Research Gap

The literature reveals a notable gap at the intersection of three domains. Research on cross-disciplinary teaching has focused on school-level subject reassignment rather than the more demanding vocational context. Self-directed learning research has examined teacher professional development but has not addressed the extreme case of qualification-free subject mastery. Technology-supported professional development research has concentrated on formal programmes with institutional infrastructure. No study has documented a teacher's trajectory from having zero formal qualification in a discipline to becoming a research-active programme leader in that discipline through self-directed, technology-assisted study. The present study addresses this gap.

§3 Methodology

3.1 Research Design

This study employed a longitudinal single-case study design (Yin, 2018). The case study approach was appropriate because the research question addressed a 'how' question about a contemporary phenomenon within its real-world context, where the boundaries between the phenomenon and context were not clearly evident. Flyvbjerg (2006) has demonstrated that the single case study, far from being merely exploratory, can generate generalisable insights when the case is strategically selected and analytically deep.

3.2 Case Selection

Teacher Z was selected based on four criteria derived from the research questions. First, ****prototypicality of the cross-disciplinary phenomenon****: Teacher Z was representative of the structural phenomenon of vocational teachers assigned to teach outside their formal qualification in China and other developing economies, making the case theoretically informative rather than merely convenient. Second, ****data completeness****: Teacher Z had maintained continuous teaching journals and system usage records from 2021 to 2025, providing an unusual depth of process data. Third, ****degree of cross-disciplinarity****: the distance between Teacher Z's formal qualifications (land resource management BSc, environmental engineering MSc) and teaching assignment (10 UAV technology courses) constituted a substantial cross-disciplinary gap, making the learning process observable. Fourth, ****ecological validity****: all data were generated in an authentic workplace setting with no intervention from the researcher.

3.3 Data Sources

Data were triangulated from four sources. **** (a) Teaching journals (2021–2025):**** Teacher Z maintained weekly reflective journals documenting learning activities, teaching problems encountered, solutions developed, and reflections on professional growth. The complete corpus comprised approximately 45,000 Chinese characters across 48 months. **** (b) Agent system usage records:**** The multi-agent support system logged all user interactions, including query types, response patterns, and task completion data. Records covered the period from system deployment in mid-2021 through mid-2025. **** (c) Institutional outputs:**** Three funded research projects obtained in 2025 (one provincial-level, two institutional-level), one copyright registration for teaching materials, and course design documents for 10 courses constituted

verifiable evidence of professional output. **(d)** Course design documents: Syllabi, lesson plans, assessment instruments, and teaching materials for 10 UAV courses developed sequentially between 2022 and 2025.

3.4 Data Analysis

Analysis proceeded in three stages. First, **timeline coding** was applied to journal and system records to establish a chronological map of developmental phases. Events were coded by type (knowledge acquisition, teaching activity, research activity, system interaction) and mapped to monthly time units. Second, **thematic analysis** (Braun & Clarke, 2006) was conducted on the journal corpus to identify recurring patterns in learning strategies, challenges, and solutions. Themes were generated inductively and then cross-referenced with the timeline map. Third, **evidence triangulation** was performed by matching themes from journals against system usage records and institutional outputs to confirm or disconfirm interpretive claims.

3.5 Ethical Considerations

Informed consent was obtained from Teacher Z for the research use of all data. Anonymisation of teacher identity and institutional affiliation was implemented throughout. Institutional approval for data use was obtained. This study did not involve vulnerable populations and posed minimal risk to participants.

3.6 Cross-Validation

In addition to the primary case, three auxiliary cases were documented as part of the broader research project: (a) Teachers S and W, two engineering instructors transitioning from solid waste engineering to smart emergency management and smart human settlements engineering respectively; and (b) Teacher H, a 61-year-old retired engineer undertaking self-directed study of traditional Chinese flute music using the same NFSC methodology. These auxiliary cases are not reported in full in this paper but are drawn upon in the Discussion (§5.2) to provide preliminary evidence of transferability. De-identified summary data for all three auxiliary cases, including timeline records and agent system usage logs, are available as Supplementary Material submitted with this manuscript.

§4 Results

4.1 Self-Study Phase (2021–2022): Knowledge Domain Mapping

The first phase of Teacher Z's development extended from mid-2021 through late 2022 and was characterised by intensive knowledge domain mapping. The teacher's prior knowledge base comprised two primary domains: land resource management (surveying, geographic information systems, spatial data analysis) and environmental engineering (monitoring technologies, impact assessment, regulatory frameworks). Neither domain included UAV-specific content, but both contained adjacent knowledge that could serve as anchoring points.

The teacher employed the NFSC approach as a self-study framework. In the Narrative phase, the teacher constructed a professional narrative that contextualised the cross-disciplinary transition as a continuity rather than a rupture: land surveying → environmental monitoring → UAV-based aerial surveying. This narrative provided coherence and motivation. In the Framework phase, the teacher systematically mapped the UAV knowledge domain, identifying sub-domains (flight principles, control systems, sensors, communications, regulations, applications) and cross-referencing them against existing knowledge to identify gaps. The agent system was used at this stage to generate structured outlines, explain unfamiliar concepts, and locate technical documentation for each sub-domain.

Journal entries from this period revealed a deliberate strategy of knowledge transfer. On 12 March 2022, Teacher Z wrote: "The GIS coordinate systems I learned as an undergraduate map directly onto drone flight path planning. The monitoring protocols from my master's research map onto drone-based environmental inspection. I am not starting from zero; I am extending existing frameworks into new territory" (translated from Chinese). This strategy of anchoring new knowledge in existing frameworks is consistent with the boundary-crossing mechanism of coordination identified by Akkerman and Bakker (2011).

Table 1 presents a condensed timeline of critical events in Teacher Z's development.

Table 1: Critical Events Timeline of Teacher Z's Development (2021–Present)

Period	Phase	Key Events
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2021 mid	Transition	Assigned to UAV programme; NFSC self-study framework adopted

| 2021 Q3-Q4 | Knowledge mapping | Agent system deployed; UAV knowledge domain structured; mapping from land management and environmental engineering domains |

| 2022 Q1-Q2 | Framework building | Technical knowledge acquisition: aerodynamics, flight control, sensor technology, civil aviation regulations |

| 2022 Q3 | Course design begins | First two course syllabi developed using agent system for structure and content generation |

| 2022 Q4 | First teaching | Initial course delivery; teaching journals document challenges and iterative refinement |

| 2023 | Sequential course development | Six additional courses designed and delivered; agent system usage shifts from knowledge acquisition to curriculum design |

| 2024 | Programme completion | All 10 UAV courses active; teaching competence externally recognised (student certification pass rates) |

| 2024 Q3 | Research initiation | Doctoral enrolment in education at [University B]; research methodology applied to teaching practice |

| 2025 Q1-Q2 | Research outputs | Three funded research projects obtained; copyright registration completed; first teaching reform manuscripts drafted |

| 2025 Q3-ongoing | Institutional embedding | Course materials standardised; mentoring of new cross-disciplinary teachers begins |

4.2 Teaching Phase (2022–2024): Agent-Assisted Curriculum Design

The second phase involved translating domain knowledge into curriculum design. Between mid-2022 and mid-2024, Teacher Z developed 10 UAV courses sequentially, with the agent system serving as a structural scaffold for curriculum design. Table 2 summarises system usage data across phases.

Table 2: Agent System Usage Data by Phase

Phase	Primary Query Type	Monthly Interactions	Output Type
Self-study (2021–2022)	Knowledge acquisition: concept explanation, domain mapping, regulatory frameworks	120–180	Structured knowledge outlines; concept maps; technical summaries

| Course design (2022–2023) | Curriculum structure: syllabus generation, lesson sequencing, learning objective formulation | 80–140 | Course syllabi; lesson plans; assessment rubrics |

| Teaching delivery (2023–2024) | Pedagogical support: teaching scenario generation, troubleshooting scenarios, assessment item creation | 60–100 | Case examples; failure scenarios for simulation; assessment items |

| Research integration (2024–2025) | Academic writing: literature search assistance, manuscript structure, proposal drafting | 40–60 | Research proposals; manuscript drafts; project reports |

The 10 courses developed cover the full UAV programme curriculum: Aviation Data Analysis and Processing, UAV Control and Trajectory Planning, Sensors and Detection Technology, Fundamentals of Mechanical Design, UAV Flight Simulation, UAV Ground Control Station, UAV Flight Control Principles and Applications, Agricultural Plant Protection UAV Technology, UAV Power Line Inspection Technology, and UAV Primary Mission Flight. Each course passed through an iterative cycle of design, delivery, reflection, and revision, with the agent system serving as a design partner throughout.

Figure 1 illustrates the knowledge domain mapping trajectory.

Figure 1: Knowledge Domain Mapping Trajectory

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Land Resource Management —————> UAV Surveying Applications



Environmental Engineering —————> UAV Environmental Monitoring



Pedagogical Practice —————> UAV Teaching Methodology



NFSC + Agent System —————> UAV Research & Scholarship

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Note: The figure shows how Teacher Z's prior knowledge domains (left) were systematically extended into UAV sub-domains (right), with the NFSC methodology and agent system serving as the bridging mechanism.

4.3 Research Integration Phase (2024–2025): From Teaching Practice to Academic Output

The third phase, from mid-2024 onward, represented the transition from teacher-as-practitioner to teacher-as-researcher. In 2024, Teacher Z enrolled in a doctoral programme in education at [University B], formalising the research dimension of professional growth. This was not the beginning of research activity—reflective journaling had been continuous since 2021—but its institutionalisation.

In 2025, Teacher Z obtained three funded research projects: a provincial-level teaching reform project on UAV programme curriculum system innovation, an institutional-level project on agent-assisted teaching models for vocational UAV education, and an institutional-level project on cross-disciplinary teacher self-study pathways. A copyright registration for the NFSC teaching methodology was also completed. These outputs constituted verifiable evidence that the self-study model not only enabled teaching competence but also supported the transition to research productivity.

4.4 Institutional Outputs

The tangible outputs of Teacher Z's trajectory are summarised in Table 3.

Table 3: Institutional Outputs as Indicators of Professional Growth (Not Direct Measures of Teaching Effectiveness)

Output Category	Item	Year	Significance
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Curriculum	10 UAV course syllabi and teaching materials	2022–2025	Complete programme coverage
Research (funded)	Provincial teaching reform project (UAV curriculum)	2025	External peer validation

| Research (funded) | Institutional project (agent-assisted UAV teaching) | 2025 |
Methodological innovation recognised |

| Research (funded) | Institutional project (cross-disciplinary self-study) | 2025 |
Professional development model formalised |

| Intellectual property | Copyright registration: NFSC + Agent teaching system | 2025 |
Transferable methodology |

| Academic enrolment | Doctoral programme in Education at [University B] | 2024 | Formal
research capacity building |

| Mentoring | Mentoring of new cross-disciplinary teachers | 2025-ongoing | Institutional
dissemination |

4.5 Knowledge Transfer Patterns

Analysis of teaching journals identified three recurrent patterns in how Teacher Z built disciplinary competence. **Pattern 1: Analogical transfer.** The teacher consistently used concepts from land resource management and environmental engineering as analogies for UAV concepts. For example, the coordinate system framework from GIS was mapped onto flight path geometry, and environmental sampling protocols were mapped onto drone-based data collection workflows. **Pattern 2: Progressive problematisation.** The teacher deliberately sought out increasingly difficult teaching tasks—beginning with simulation-based courses before progressing to equipment-intensive courses—creating a self-imposed zone of proximal development. **Pattern 3: Reflective iteration.** Each teaching cycle was followed by a structured reflection documented in the journal, identifying specific knowledge gaps, pedagogical failures, and revision strategies for the next iteration.

§5 Discussion

5.1 RQ1: How the Self-Directed Model Supports Cross-Disciplinary Competence Development

The findings demonstrate that a self-directed, technology-assisted learning model can support cross-disciplinary teaching competence development under specific conditions. Teacher Z's trajectory exhibits all four boundary-crossing mechanisms identified by Akkerman and Bakker (2011). **Identification** occurred through the narrative construction of professional continuity. **Coordination** was accomplished through systematic knowledge domain mapping and the

intentional use of analogical transfer. **Reflection** was embedded in the weekly journaling practice. **Transformation** materialised in the integration of teaching, research, and continued learning into an ongoing professional identity.

The technology support system played a critical but circumscribed role. It was not a learning management system that delivered content, nor an intelligent tutoring system that tracked competency. Rather, it functioned as an always-available knowledge scaffold—a responsive information resource that the teacher could query at any point in the learning process. This aligns with Eraut's (2004) characterisation of informal workplace learning, in which most learning occurs during work activity rather than in dedicated training time. The system made knowledge accessible at the point of need, eliminating the temporal gap between encountering a knowledge gap and accessing the information to fill it.

The NFSC methodology provided the structure that Brookfield (1986) identified as essential for effective self-directed learning. The four components—Narrative (constructing a coherent professional story), Framework (systematic knowledge mapping), Scene (contextualised scenario-based learning), and Creation (producing tangible outputs)—constituted a self-study protocol that substituted for the curriculum structure of formal training. The NFSC approach draws on established pedagogical principles including narrative pedagogy (Bruner, 1986), constructive alignment (Biggs, 1996), and scenario-based learning, adapted here for self-directed knowledge acquisition. This finding extends the work of Verster et al. (2024) by demonstrating that self-directed professional development can address not only curriculum practice enhancement but also foundational disciplinary knowledge acquisition. We acknowledge the potential for confirmation bias given the research team's involvement in developing the NFSC methodology. To mitigate this, we triangulated self-report data (teaching journals) with independent evidence sources—funded project awards and copyright registration—that were adjudicated by external reviewers unaware of this study.

5.2 RQ2: Conditions for Cross-Disciplinary Transfer

The three auxiliary cases provide preliminary evidence for transferability conditions. Teachers S and W, both originally from solid waste engineering, applied the same NFSC methodology to transition into smart emergency management and smart human settlements engineering respectively. Their transitions were completed within 12–18 months, generating new course

materials for undergraduate programmes (four courses for Teacher W, three for Teacher S). Teacher H, a 61-year-old retired engineer with no musical background, used the methodology to acquire competence in traditional Chinese flute music, developing a structured self-learning programme termed the Bamboo-Music-Wellness system, which subsequently received copyright registration and was formalised into a community teaching programme. Across all four cases, three conditions for successful transfer emerged.

****First, the existence of an anchoring knowledge domain.**** In every case, the teacher possessed a pre-existing knowledge structure—land management and environmental engineering for Teacher Z, solid waste engineering for Teachers S and W, engineering problem-solving for Teacher H—that provided conceptual frameworks onto which new knowledge could be mapped. The agent system facilitated this mapping by making the structure of the target domain explicit and highlighting connections to the anchoring domain.

****Second, a structured self-study protocol.**** The NFSC methodology provided a replicable sequence of cognitive operations: narrate, map, simulate, produce. This protocol served the same function as the structured reflection activities that Verster et al. (2024) found essential for self-directed professional development, but extended it from reflection-on-practice to structured knowledge acquisition.

****Third, institutional permission for self-directed experimentation.**** In all four cases, the teachers operated within institutional environments that allowed—or at least did not actively obstruct—self-directed learning activities. This condition aligns with the findings of Rasool and Naidoo (2024) regarding the importance of institutional enabling conditions for technology-supported professional development.

5.3 Policy Implications for Developing Countries

5.3.1 Cost Accessibility and Infrastructure

The model described in this paper operates at zero marginal cost to institutions. The agent system requires only a standard internet-connected computing device and a freely available large language model interface. Deployment involves no server infrastructure, no licensing fees, and no specialised technical personnel. This addresses the infrastructure barriers that Chuene and Teane (2024) identified as critical constraints on curriculum implementation in resource-limited

South African schools. For education departments in developing economies facing the simultaneous pressures of curriculum expansion, teacher shortages, and fiscal constraint, a zero-cost teacher self-development infrastructure represents a pragmatic policy option.

5.3.2 Cross-Domain Replicability

The range of disciplinary transitions documented—from solid waste engineering to unmanned aerial vehicles, from environmental science to smart engineering, from engineering to traditional music—suggests that the model transcends specific disciplinary boundaries. This cross-domain applicability is significant for policy-makers because it means a single methodology can address teacher shortages across multiple programme areas simultaneously, rather than requiring discipline-specific training interventions.

5.3.3 Institutional Embedding

A critical policy question is whether self-directed models can be integrated into existing teacher continuing education frameworks. Sepadi and Themane (2025) demonstrated that short learning programmes can build capacity for inclusive education among South African teachers, but noted that sustained institutional support is required for programmes to translate into practice change. For the model described here, institutional embedding would require three elements: (a) official recognition of self-directed learning activities as legitimate continuing professional development, (b) incorporation of the NFSC methodology into induction programmes for cross-disciplinary teachers, and (c) establishment of peer support networks for cross-disciplinary teachers using shared technology platforms.

5.3.4 BRICS and South-South Cooperation Framework

The relevance of this model to South African education policy extends beyond domestic application. Within the BRICS cooperation framework (Brazil, Russia, India, China, South Africa), teacher training has been identified as a priority area for collaborative action (BRICS Ministers of Education, 2023). The model's zero-cost architecture, cross-domain applicability, and independence from proprietary infrastructure make it suitable for South-South knowledge transfer. In contexts such as South Africa's technical and vocational education and training (TVET) college sector, where lecturer qualifications remain a pressing concern (Naidoo, 2026), a

methodology that enables self-directed competence development without external training resources addresses a direct policy need.

5.4 Limitations

Several limitations constrain the interpretation of this study's findings. First, it is a single-case study ($N = 1$), and while Flyvbjerg (2006) has established that single cases can yield generalisable insights, the external validity of the findings remains to be established through replication. Second, there is no control condition; the study cannot determine whether Teacher Z would have achieved comparable or even superior results through alternative approaches, nor can it isolate the specific contribution of the technology component from the NFSC methodology and the teacher's individual characteristics. Third, the data rely substantially on self-report through teaching journals, introducing potential self-report bias (Patton, 2015). Fourth, the study does not include student learning outcome data that would permit evaluation of teaching effectiveness beyond institutional indicators such as funded research projects. Fifth, Teacher Z's particular combination of strong pre-existing analytical skills (from two previous degree programmes), high motivation (doctoral enrolment), and access to computing infrastructure may limit the generalisability of findings to teachers with different personal and contextual characteristics.

§6 Conclusion

This study set out to examine how a self-directed, technology-assisted learning model supports cross-disciplinary teaching competence development in vocational education, and to identify conditions that enable its transfer across disciplines. The findings from the longitudinal case of Teacher Z demonstrate that such a model can support the full trajectory from disciplinary novice to research-active programme leader, given the presence of an anchoring knowledge domain, a structured self-study protocol, and institutional permission for experimentation.

Three contributions emerge from this research. The first is empirical: the documentation of a complete cross-disciplinary teacher development trajectory, from initial knowledge mapping through curriculum design to scholarly output, fills a documented gap in the teacher professional development literature. The second is methodological: the NFSC self-study protocol, adapted from classroom pedagogy to self-directed learning, offers a replicable structure for cross-

disciplinary knowledge acquisition that extends existing models of self-directed professional development (Verster et al., 2024). The third is policy-oriented: the zero-cost architecture of the technology support system, combined with evidence of cross-domain replicability, provides a practical option for education systems in developing economies facing structural teacher shortages.

These contributions must be weighed against the limitations of a single-case design. The model is not a replacement for formal teacher education. A teacher with specialist disciplinary training will, all else being equal, arguably bring deeper and more integrated knowledge to the classroom than one who is self-taught. The model is positioned as a pragmatic bridge for contexts where formal training is inaccessible—contexts that are, unfortunately, prevalent in the developing-country vocational education systems this journal serves.

Future research directions include: multi-case comparison studies to establish the boundary conditions of the model; quantitative studies incorporating student learning outcome measures to evaluate teaching effectiveness; cross-institutional replication studies in different national and disciplinary contexts; and design-based research to optimise the technology support system for specific disciplinary transitions. The auxiliary cases reported briefly in this paper suggest that the model's transferability across disciplines and life stages warrants systematic investigation. In a global education landscape characterised by persistent teacher shortages, resource constraints, and rapidly evolving curricula, self-directed technology-assisted models of teacher professional growth merit both scholarly attention and cautious policy consideration.

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- ****Data availability:**** Data are available from the corresponding author upon reasonable request, subject to institutional privacy constraints.

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